Graduation Project Proposal

Project Information		
Project Title	**Department/Faculty/University**	**Project Field/Discipline**
Donation and Emergency Aid App	**Computer science/Future University**	
Advisors' Names	**Advisors' Mobile Numbers**	**Advisors' Email Addresses**
TA/Debaj		**debaj.elmaghraby@fue.edu.eg**
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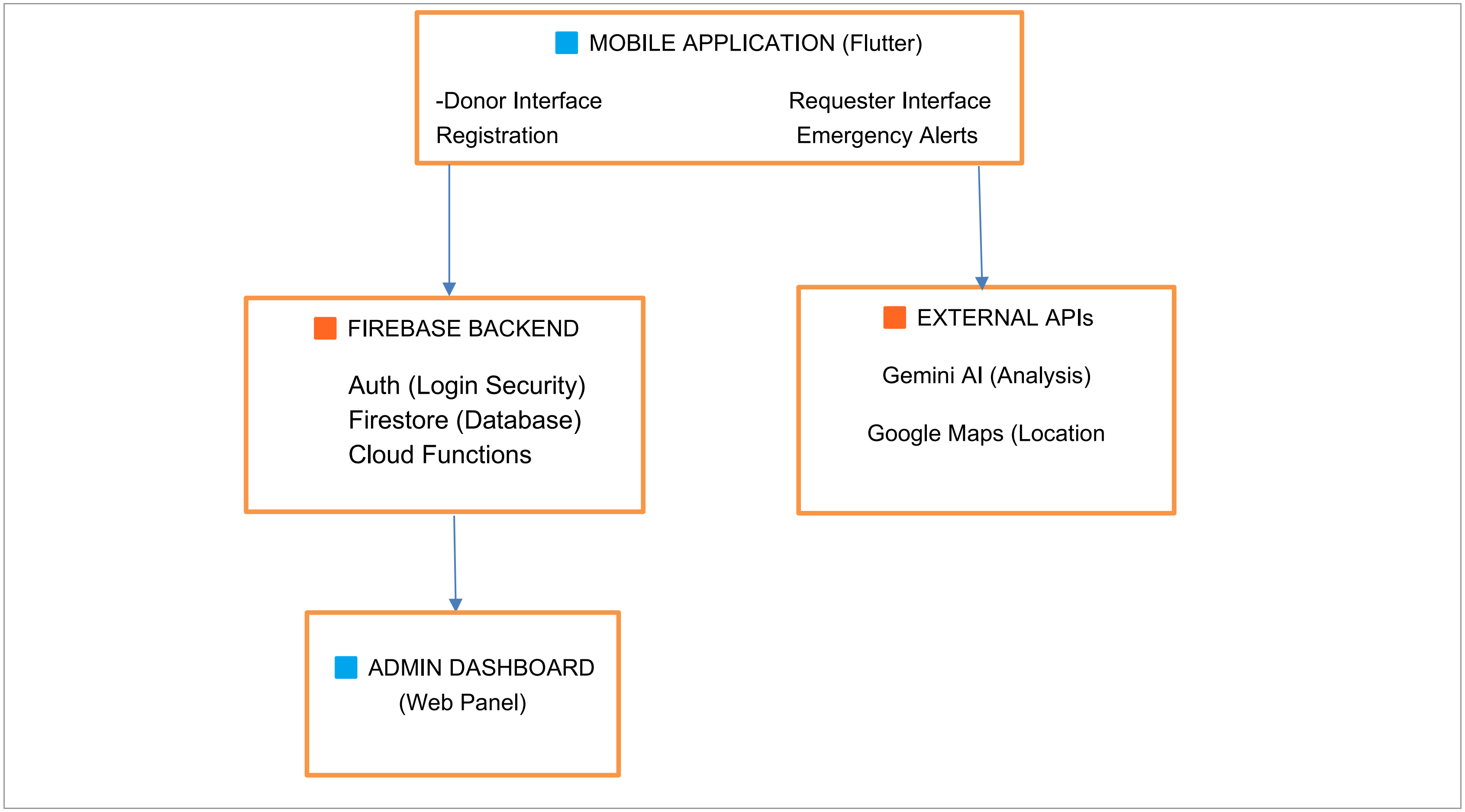
Is this project part of a mega-project at the same institution? ☐Yes ☐No (If yes, please submit all proposals together.)
If the project is sponsored by or initiated by an ICT company, please state its name:

Motivation
Please write why you chose this project idea, explaining clearly the problem that the project is addressing
we chose this project because many people face serious problems getting quick help during emergencies such as accidents, fires, or medical situations. Often, donations and support take too long to reach those who need them. There is no simple or fast way to connect people who want to help with people who need help in real time. Our motivation is to use technology to make this process faster and more effective. By creating one smart platform, we can bring donors, organizations, and people in need together in one place. This can help save lives, reduce delays, and build a stronger community where people support each other easily.
Why do you think your project should be funded? for which the applicants write in a few lines where the help statement should be "Explain in no more than 3 lines the new and innovative aspects in your project that make it worthy of funding."

This project is innovative because it combines mobile technology, real-time location, and AI to speed up emergency donations. It provides a fast and smart way to help people in need. With funding, it can grow and make a real impact in saving lives.
Block Diagram
Please insert the project detailed block diagram below, (Please highlight the parts that will be implemented in different colors than the parts that will be purchased)


MOBILE APPLICATION (Flutter)
-Donor Interface
Registration
Requester Interface
Emergency Alerts
FIREBASE BACKEND
Auth (Login Security)
Firestore (Database)
Cloud Functions
EXTERNAL APIs
Gemini AI (Analysis)
Google Maps (Location
ADMIN DASHBOARD
(Web Panel)

Prototype Description and Specifications

Please note that ITAC only funds projects that result in a prototype. Include a clear description of how the prototype will operate, explaining a scenario/use case of the operation. Also include the performance metrics you target in the prototype.

1. Request Initiation: A user (Requester) logs in and taps "Emergency Request." They select "Blood Donation," specify the blood type, and the app automatically captures their GPS location.

2. AI Matching & Notification: The system's AI analyzes the request urgency and scans the database for registered donors with the matching blood type who are currently within a 5km radius.

3. Real-Time Alert: Identified donors receive an instant push notification: "Urgent O+ Blood needed 2km away."

4. Response: A donor accepts the request. The app generates a route on the map connecting the donor to the hospital/requester.

5. Completion: Upon arrival, the donor confirms the action via a QR code scan at the location, and the system updates the request status to "Fulfilled."

Target Performance Metrics:

- Notification Latency: < 2 seconds from request creation to donor alert.
- Location Accuracy: GPS tracking precise within 10–15 meters.
- AI Processing Speed: Matching algorithm execution < 1 second.
- Concurrency: Support at least 50 simultaneous active requests during the demo.

Project Plan

Please define the approach and phases to deliver the intended project outcome.

Project Plan

Phase 1: Requirement Analysis & UI/UX Design

- Define detailed functional requirements for donors and organizations.
- Design wireframes and high-fidelity prototypes for the mobile app.
- Outcome: Finalized SRS document and UI designs.

Phase 2: Core Development & Database Setup

- Set up the Cloud Database (Users, Requests, Inventory).
- Develop Authentication (Login/Sign-up) and User Profile management.
- Implement the "Create Request" functionality.
- Outcome: Functional Alpha version with basic CRUD operations.

Phase 3: Geolocation & AI Integration

- Integrate Map APIs for real-time location tracking.
- Develop the AI logic for sorting and prioritizing emergency cases.
- Implement Push Notification services.
- Outcome: Beta version with location services and smart alerts.

Phase 4: Testing & Refinement

- Unit testing for individual modules.
- Integration testing (Front-end with Backend).
- User Acceptance Testing (UAT) with sample scenarios.
- Outcome: Stable prototype ready for the graduation exhibition.

Prototype Prospects

List the Egyptian ICT companies that may be interested in the developed prototype and the end-users/customers (name the specific class of individuals, governmental agencies, ministries … etc. that will benefit from the prototype)

List of ICT Companies:
Fawry / e-Finance: Potential partners for integrating donation payment gateways.

Orange / Vodafone Egypt: Interested in the location-based service infrastructure or CSR (Corporate Social Responsibility) sponsorship.

Software Houses (e.g., ITWorx, Valeo): May be interested in the talent behind the AI/Location integration.

Potential End-Users/ Consumers:
NGOs & Charities: Egyptian Red Crescent, Resala Charity Association, Misr El Kheir Foundation (to manage volunteer logistics).

Governmental Agencies: Ministry of Social Solidarity (for coordinating aid), Ministry of Health (for blood donation emergencies).
General Public: Any citizen willing to donate or in need of emergency aid.

** Please fill in the budget form and upload both the proposal & budget forms to the student application form.